

DAVID YUCHEN WANG

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As a current AI Algorithm Engineer at TikTok, I hold a Master's Degree in AI from the Nation University of Singapore, and a Bachelor's Degree in Engineering Physics from the University of British Columbia, Canada. I have 6+ years of work experience in deployment and research of Multi-modal LLMs, generative AI, reinforcement learning, computer vision, and deep learning. I am a quick learner, an excellent team player, and possess strong work ethic. I am fluent in English and Mandarin, currently hold a Canadian citizenship and open to exploring all opportunities in AI.

EDUCATION

National University of Singapore

Aug 2023 - Dec 2024

Master of Computing - Specialization in Artificial Intelligence

Thesis: Enhanced Deep Neural Network for Video Copy Detection and Localization

Courses: Probabilistic Graphical Models, Deep Learning, Distributed Systems, AI Decision making, 3D Computer Vision, Natural Language Processing, Big Data Systems

University of British Columbia, Canada

Sep 2018 - May 2023

Bachelor of Applied Science - Major in Engineering Physics, Minor in Commerce.

- UBC Presidential Scholars Award Recipient, Dean's Honors List

WORK EXPERIENCE

AI Algorithm Engineer

Jun 2024 - Present

Content Intelligence Team, TikTok Singapore - <https://www.tiktok.com>

- Led the end-to-end development and deployment of **multimodal (video/audio/text)** moderation models in short video, livestream, and creator-rewards product lines globally across 8 regions, serving stable online traffic of over **5000 QPS**.
- Advanced multimodal model architectures and data pipelines through **large-scale (million level) dataset curation**, MLLM-based data generation and distillation, attention-based fusion, and omni-embedding research, consistently exceeding targets on model precision and recall, achieving near-MLLM performance with $\sim 20\times$ smaller models, and co-authoring 3 papers at KDD 2026 and 3 patents.
- Scaled production ML systems and infrastructure by **optimizing inference** (increasing model throughput by over 3x), building internal model-serving and self-service platforms (reaching 0 $\rightarrow$ 3,000 weekly users in 1 year), and delivering over **\$40M USD in business savings** over 2026.

Edge AI Developer

Aug 2023 - May 2024

Pensees Singapore - <https://www.pensees.sg>

- Adapted state-of-the-art generative **diffusion pipelines** for iOS using SwiftUI and Bazel, leveraging enhancements with **CoreML**, **MPS**, and **Metal** to achieve native on-device functionality with platform-specific optimization.
- Spearheaded the project development, integrating insights from weekly analyses of cutting-edge **research papers** to enhance team expertise and guide project trajectory, built and launched an **AI-drawing app** within 2 months.
- Acquired and applied expertise in various cutting-edge methods for use and deployment of large-scale image diffusion models, including **controlNet**, **LoRAs**, **consistency models**, and **NLP prompting**.

Machine Learning Research Assistant

May 2021 - Aug 2023

TRIUMF - Canada's Particle Accelerator Centre - <https://www.triumf.ca>

- Developed **Bayesian Optimization** model to optimize particle beamlines and boost speed by 2400% and accuracy by 120% compared to human operators.
- Designed efficient physics simulations for use on policy gradient **reinforcement learning** models. Integrated first AI-controlled interface on particle accelerators.
- Published experimental findings as first author in paper *Accelerator Tuning with Deep Reinforcement Learning* and gave video and poster presentation at **NeurIPS 2021** workshop.

Machine Learning Engineer

May 2022 - Sep 2022

Yakoa.io - Web3 Startup - <https://www.yakoa.io>

- Implemented **image segmentation** framework in PyTorch from research papers to detect fraudulent features in NFT images with high accuracy.
- Deployed **self-supervised classification** models on AWS instances and fine-tuned models on a dataset of 8 million images, improving model run-times by over 300%.
- Employed **statistical analysis** of latent space of self-supervised models. Optimized hyperparameters and visualized results using Weights & Biases, leading to 150% improvement in validation accuracy.

AI Research Intern

Jan 2020 - May 2020

Huawei Technologies Canada - Vancouver Big Data Lab

- Enhanced **data-preprocessing** speeds for large image datasets by 300% through designing scripts in Python and Bash.
- Boosted team productivity by 500% through configuring **custom environments** in Docker to allow parallelization of model training through cloud GPU clusters.
- Fine-tuned **deep-learning** models for **image classification** and **object detection** in TensorFlow and PyTorch. Organized documentation and presented findings to team, leading to 120% improvements on model accuracy.

Captain and Navigation Sub-Team Lead

Sep 2019 - May 2023

UBC AgroBot - Student Engineering Design Team - <https://ubcagrobot.com>

- Devised project roadmaps, established a 2-year budgeting timeline, and utilized **Agile methodology** to manage a team of 70 members across 6 sub-teams to bring robot to the 2023 METRICS ACRE international competition.
- Led a team of 8 members to integrate software with hardware systems onboard robot. Interfaced with camera, lidar, and gyro sensors and utilized **computer vision** and PID control to achieve **autonomous navigation** through crop fields.

SKILLS

Programming languages: Python, Java, Swift, MATLAB, C++, C, C#, Julia, R, HTML, CSS

Libraries: PyTorch, Pandas, TensorFlow, Keras, OpenCV, Numpy, SciPy, Matplotlib, Weights & Biases

Frameworks: SQL, PySpark, HDFS, Linux, Bash, Bazel, Xcode, Slurm, ROS, Gazebo, AWS, GIT, Docker, Conda, Arduino

PUBLICATIONS

MLT-Dedup:

Efficient Large-Scale Online Video Deduplication via Multi-Level Representations and Spatial-Temporal Matching

<https://arxiv.org/abs/2606.12215>

KDD 2026 - ACM SIGKDD Conference on Knowledge Discovery and Data Mining

MatchLM2Lite: A Scalable MLLM-to-Lite Framework for Reproduced Content Identification

<https://arxiv.org/abs/2606.14786>

KDD 2026 - ACM SIGKDD Conference on Knowledge Discovery and Data Mining

Dynamic Content Moderation in Livestreams: Combining Supervised Classification with MLLM-Boosted Similarity Matching

<https://arxiv.org/abs/2512.03553>

KDD 2026 - ACM SIGKDD Conference on Knowledge Discovery and Data Mining

Accelerator Tuning with Deep Reinforcement Learning

https://ml4physicalsciences.github.io/2021/files/NeurIPS_ML4PS_2021_125.pdf

NeurIPS 2021 - Workshop for Machine Learning and the Physical Sciences

LEADERSHIP EXPERIENCE

Captain - UBC AgroBot Design Team (Sep. 2022 - May 2023)

Grad Year Representative - UBC Engineering Students Council (Sep. 2022 - May 2023)

Teaching Assistant - UBC (2020 - 2023) across 3 courses in ML and Physics

PERSONAL INTERESTS

Hiking, Photography, Piano, Badminton, Running